

CLOUDWALK AML/CTF CASE

Reasoning & Methodology

How the case was solved — the thinking behind the findings, not the findings themselves.

Companion deck to the Board / Examiner results briefing.

Every slide reads: **Problem** → **Hypotheses** → **Test** → **Decision** → **Alternatives rejected**

HOW I FRAMED THE PROBLEM

Why treat this as a real BSA/AML program audit — not a spreadsheet exercise?

THE PROBLEM A synthetic workbook can be "solved" as data trivia. The framing chosen determines whether the answer survives an examiner. Every finding must carry an action, an owner, and a legal authority — or it is not finished.

HYPOTHESIS A · DATA PUZZLE

Count the anomalies and hand back a list.

✗ Cosmetically complete — but produces facts with no dispositions.

HYPOTHESIS B · PROGRAM LIFECYCLE

An end-to-end program lifecycle with dispositions.

✓ Answers the examiner: "what did you do about it?"

DECISION · ADOPT THE REGULATOR LIFECYCLE

Each finding maps to **file / monitor / close**, with owner + authority. **Rejected:** pure spreadsheet framing — facts with no "reason to suspect → act" pathway; fails the moment an examiner asks for the disposition.

THE FIRST CRITICAL DECISION

Validate the population before analysing anything

THE PROBLEM openpyxl's ws.max_row returns 1,000 — it counts blank trailing rows. Using ~999 as the denominator is the trap.



HYPOTHESES (a) 999 rows ≈ population — analyse directly. (b) Reconcile the real population by unique ID before any ratio is computed.

TEST • DECISION

Reconcile IDs → **501 / 100 / 630**. Bake an `assert len==501` into the script; mandate a Section-0 population check before any percentage. **Rejected:** the 999 denominator — it understates every downstream ratio by ~10×, corrupting the analysis invisibly.

SCOPE OF THE EDD DELIVERABLE

Deliver the full prioritised population — not a sample of five

THE PROBLEM The minimal ask was a short EDD sample (≈5 illustrative customers). Is a sample enough to defend?

SAMPLE OF 5 · REJECTED

Cosmetically complete, operationally useless. Leaves **~237 real exposures** unmanaged and cannot feed a capacity or remediation model.

FULL POPULATION · RANKED

242 / 501 customers trigger ≥1 EDD criterion

(214 on the volume/KYC-mismatch criterion alone). Delivered ranked by urgency, with deadline + documents required — executable, not illustrative.

COST / BENEFIT THAT SETTLED IT

Marginal compute cost to score all 501; large defensibility gain. A ranked full population is the only output that supports a real remediation queue, an FTE capacity model, and an examiner walkthrough. Sampling would trade all of that for nothing saved.

THE C8888 DILEMMA

Why escalate for adjudication — not file, and not archive?

THE PROBLEM One customer: sanctions_score 0.84, Iranian-passport provenance, risk_rating only "medium". Three exits were on the table.

HYPOTHESIS A · CONFIRMED MATCH

0.84 is a real match → file the SAR now, apply IRAN-2025-A002.

HYPOTHESIS B · SCREENING ARTIFACT

0.84 is anglicised-name vs. Farsi SDN noise → possibly a false positive.

TEST · FINCEN STANDARD A score is a signal to investigate, not evidence of a reportable transaction (FinCEN SAR FAQs, Oct 9 2025). Neither A nor B can be asserted yet.

DECISION

Escalate to OFAC adjudication; upgrade to **high / enhanced**; precautionary hold. Keep IRAN-2025-A002 contingent on a confirmed match. **Rejected:** "file now" (premature) and "archive" (ignores a subcalibrated rating over a live Iran nexus). Confirm before you assert.

THE OFAC 50% RULE

One rule, two arithmetics — the moment "3 merchants" proved incomplete

THE PROBLEM How many merchants are potentially blocked entities under the 50% Rule? The intuitive count was three.

MENTAL MODEL 1 · SINGLE OWNER

One Iranian UBO holding $\geq 50\%$.

3 merchants

M3001 (87.35%) · M3012 (57.03%) · M3056 (59.17%, PEP)

Correct — but only under the single-owner reading.

MENTAL MODEL 2 · AGGREGATION

Ownership is aggregated — *"one or more blocked persons."*

+1 merchant

M3098 — two owners summing $\geq 50\%$ (18.99% + 36.42%)

The complete answer needs both arithmetics.

DECISION · THE TRAP AVOIDED

Report **4 merchants** (3 single-owner + 1 aggregate). Frame the near-zero automated sanctions scores on Iran-domiciled UBOs (**0.11 / 0.01 / 0.00**) as a calibration failure, not a clearance. Rejected: stopping at "3," which reads the rule with only one of its two arithmetics.

PRIORITISING THE PRIMARY SAR

Which of many candidate events becomes the SAR?

THE PROBLEM The dataset holds many patterns. Only some rise to a filing — the ranking criterion decides which.

2025-10-26 · one device dev_shared_4829 · one IP 45.129.55.210 · merchant M3061 (MCC 4829)

12 transactions · fixed 6-minute intervals · **\$9,664.92** in ≈66 minutes

HYPOTHESIS A · RANK BY SIZE

The biggest dollar number wins.

HYPOTHESIS B · RANK BY COORDINATION

Corroborated by independent signals.

TEST · DECISION

Two independent infrastructure signals — **device and IP** — identify the same 12-account set; timing is metronomic, not human. File as the primary SAR. **Rejected:** size-first ranking (a lone large txn lacks the signature); merchant co-visitation (expected by chance across 100 merchants).

A REGULATORY JUDGEMENT CALL

Why "Structuring" was considered and struck from the SAR

THE PROBLEM Sub-\$1k transactions at high-risk MCCs look like structuring. Should Form 111 Item 32 be selected?

FORM 111 · PART II SELECTIONS

- ✓ 36 Money laundering
- ✓ 38 Suspicious use of multiple accounts
- ✓ 42 Cyber event (shared device / IP)
- ✗ **32 Structuring**

THE INTENT TEST · FINCEN FAQs (OCT 9 2025)

Structuring turns on **intent to evade a reporting threshold**. Values \$469.87–\$1,140.74 do not cluster against the \$10,000 CTR line, and no facts show evasion intent. The theory fails on the facts.

DECISION

Do not select Item 32. Characterise as **coordinated multi-account control / layering** — Items 36, 38, 42. **Rejected:** labelling it structuring — a factually wrong theory an examiner would dismantle. Suspicion arises from control and absent lawful purpose, not dollar proximity.

DESIGNING DETECTION TO BE FALSIFIABLE

The experiment I threw away — and what it taught the real rule

THE PROBLEM How do you detect one device controlling many accounts, without drowning in false positives?

DISCARDED · LIFETIME REUSE COUNT

~669 **devices** flagged (of 5,159 shared) — almost all benign: households, kiosks. The rule failed its own false-positive test.

ADOPTED · 60-MIN SLIDING WINDOW

Self-join anchor: each row anchors a 60-min window over the same device;
DISTINCT ON dedups overlaps. Threshold ≥ 3 = **alert**, ≥ 5 = **hard-block**.

THE PRINCIPLE IT PRODUCED

Chose the burst-in-window rule: the confirmed extreme case (**12 customers on one device at MCC 4829**) sits far beyond both thresholds, so a real ring still trips it. A rule that flags almost everything has proven nothing — I keep the discarded experiment in the record, not just the fix.

WHEN A PATTERN IS TOO COMMON TO BE A SIGNAL

Why cash-in → cash-out needed a method, not a count

THE PROBLEM The ATM (6011) → remittance (4829) pattern matches ~97.6% of customers. A pattern that universal is not a signal.

97.6%

of customers do both — counting alerts on nearly everyone

STANDARD TIER · ≤14 days, ratio ≥80%

→ **3 customers** (investigator triage, wide net)

HIGH_CONFIDENCE · ≤72h

→ **exactly 1 (C12261)** KPI / board reportable

tighten to same-session proximity →

DECISION

One query, two tiers: a wide net for investigators and a tight net for reportable metrics. Every high-confidence row is also a standard row; only the one validated case is cited as confirmed. **Rejected:** a single occurrence-count — the constraint (amount linkage + time proximity) is what turns a universal behaviour into a signal.

REASONING ACROSS SECTIONS, NOT WITHIN ONE

How "GO — with conditions" came from Section 1, not Section 5

THE PROBLEM Is the new product (instant OCT payouts + crypto on-ramp) launchable? Inherent risk is 5/5.

Subcalibrated Iran / VE ratings	2 missing MSB registrations	OFAC 50% Rule undetermined	Device / card-test detection not live
---	---------------------------------------	--------------------------------------	---

4 open enterprise gaps (Section 1 risk matrix) — the product's residual-3 target holds only if these close first.

GAPS OPEN → EFFECTIVE RESIDUAL 4 (HIGH)

Above appetite → No-Go.

GAPS CLOSED + CONTROLS VALIDATED → RESIDUAL 3

Within appetite → launch, phased pilot.

DECISION · GO — WITH CONDITIONS

Rejected: assessing the product in isolation. **1,687 MCC 6051 transactions** already show sub-threshold structuring — launching amplifying products over open gaps would re-elevate effective residual to Critical.

AUDITING MY OWN WORK

Why re-open the source files with a different model?

THE PROBLEM "Done" is not the same as "correct." Self-review inherits the author's blind spots.

INDEPENDENT QA · 17 FINDINGS, EACH RE-VERIFIED AGAINST REAL DATA

9

confirmed correct

3

real but inflated in scope

5

false / mischaracterised

HYPOTHESES (a) Trust the intermediate summaries — they say it's fixed. (b) Re-open dataset, scripts & SQL byte-for-byte, different model, across 7 QA sessions.

DECISION · SUCCESS CRITERION

Every CRITICAL confirmed finding corrected and re-validated before the package is "Board-ready." The log records intent; only a reopened file is proof. Trusting summaries would have shipped a "corrected-but-not-persisted" defect — which recurred twice (the .docx-as-markdown bug, and corrections never merged into the base files).

HONESTY OVER A COMFORTABLE ANSWER

The T10 rule: document a real gap, don't fabricate a hit

THE PROBLEM The self-merchant (T10) rule produced 175 name-only collisions — pure noise. What do you report?

CONFIRMED	CORROBORATED	NAME_ONLY
population zero reachable under the current schema — tax_id / device / email are not usable identity keys today	secondary signal present — retained for future schema	175 collisions — removed from the actionable population

DECISION

Report **"population zero reachable"** as a documented schema gap. Recomputed independently in Python: 0 CONFIRMED, matching the SQL. Removing name-only kills ~21.7 noise alerts/month. **Rejected:** seeding fictional data to manufacture a CONFIRMED hit — a true zero, documented, beats a fabricated one an examiner could not reproduce.

SYNTHESIS

Three principles that governed every decision

01

Never confirm before you adjudicate.

A score, a UBO's country, a pattern — each is a reason to investigate, not a conclusion. (C88888's 0.84; the near-zero Iran scores; the SAR basis test.)

02

Never treat a universal pattern as a standalone signal.

If it matches almost everyone — cash-in/out at 97.6%, lifetime device reuse at 669 — it needs a constraint, not an alert.

03

Never accept "done" without independent re-verification.

A different model, reopened source files, CRITICAL-first. The log is intent; the file is proof. Two "corrected-not-persisted" defects were caught this way.